Notes for IE 33500 - Operations Research - Optimization

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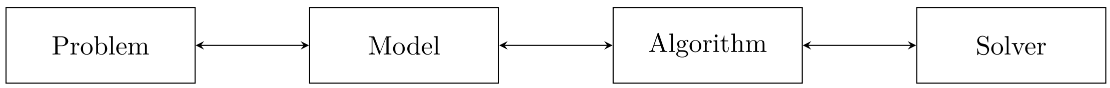
Contents

Course Description

Introduction to deterministic optimization modeling and algorithms in operations research. Emphasis on formulation and solution of linear programs, networks flows, and integer programs. Typically offered Fall Spring.

Introduction to Optimization

Optimization seeks to find the "best" solution out of all possible solutions, for some measure of "best". To do this we must *model* the problem mathematically and implement some *algorithm* to find the numerical solution to the model. *Solvers* assist us in implementing the algorithm when it's infeasible to do by hand.



Linear Programming Modeling

Consider the following problem statement.

Two Crude Petroleum (TCP) runs a small refinery on the Texas coast. The refinery distills crude petroleum from two sources, Country A and Country B, into three main products: gasoline, jet fuel, and lubricants. The two crudes differ in chemical composition and thus yield different product mixes. Each barrel of Country A crude yields 0.3 barrel of gasoline, 0.4 barrel of jet fuel, 0.2 barrel of lubricants, and 0.1 barrel of waste. On the other hand, each barrel of Country B crude yields 0.4 barrel of gasoline, 0.2 barrel of jet fuel, 0.3 barrel of lubricants, and 0.1 barrel of waste. The crudes also differ in cost and availability. TCP can purchase up to 9000 barrels per day from Country A at $20 per barrel. Up to 6000 barrels per day of Country B petroleum are also available at the lower cost of $15 per barrel because of the shorter transportation distance. TCP's contracts with independent distributors require it to produce 2000 barrels per day of gasoline, 1500 barrels per day of jet fuel, and 500 barrels per day of lubricants. How can these requirements be met most cost effectively?

Given the problem, we must now model it mathematically to solve. There are three components of a model: the *decision variables*, the *objective function*, and the *constraints*.

Decision variables In this example the only decision variables are the quantity of barrels to purchase from Country A, x_1, and from Country B, x_2.

Objective function Minimize daily cost.

Constraints

- Gasoline requirement
- Jet fuel requirement
- Lubricant requirement
- Country A availability
- Country B availability
- We must purchase natural numbers of barrels, not negative barrels, not fractional barrels, not imaginary numbers of barrels. However, for the purposes of this problem, let us impose the weaker constraint of non-negativity.

Expressed mathematically, if x_1 is the amount of Country A crude refined per day in 10^3 barrels and x_2 is amount of Country B crude refined per day in 10^3 barrels, minimize the daily cost $\gamma = 20x_1 + 15x_2$ subject to the constraints:

$$0.3x_1 + 0.4x_2 \geq 2 \tag{1}$$
$$0.4x_1 + 0.2x_2 \geq 1.5 \tag{2}$$
$$0.2x_1 + 0.3x_2 \geq 0.5 \tag{3}$$
$$x_1 \leq 9 \tag{4}$$
$$x_2 \leq 6 \tag{5}$$
$$x_1, x_2 \geq 0 \tag{6}$$

Note that that we may produce more than the demand; while this is suboptimal, if we set these constraints as equalities we may have a system of three equations with two unknowns and thus overconstraint ourselves and not be able to find a solution.

This is the *compact form* of the model, which we may now use an algorithm to solve. In this case, we can use the graphical method, which works if you have exactly two decision variables. First, we identify the feasible region.

1. Plot all constraint lines at equality
2. Determine the feasible sides of the constraint lines
3. The feasible region must satisfy all constraints

Next, find the improving direction of the objective function.

1. Arbitrarily plot two objective function lines
2. Find the direction of improvement

Finally, eyeball the optimal solution.

1. Focus on corners of the feasible solution region
2. Compute the numerical value from two crossing constraint lines
3. There may be multiple optimal solutions

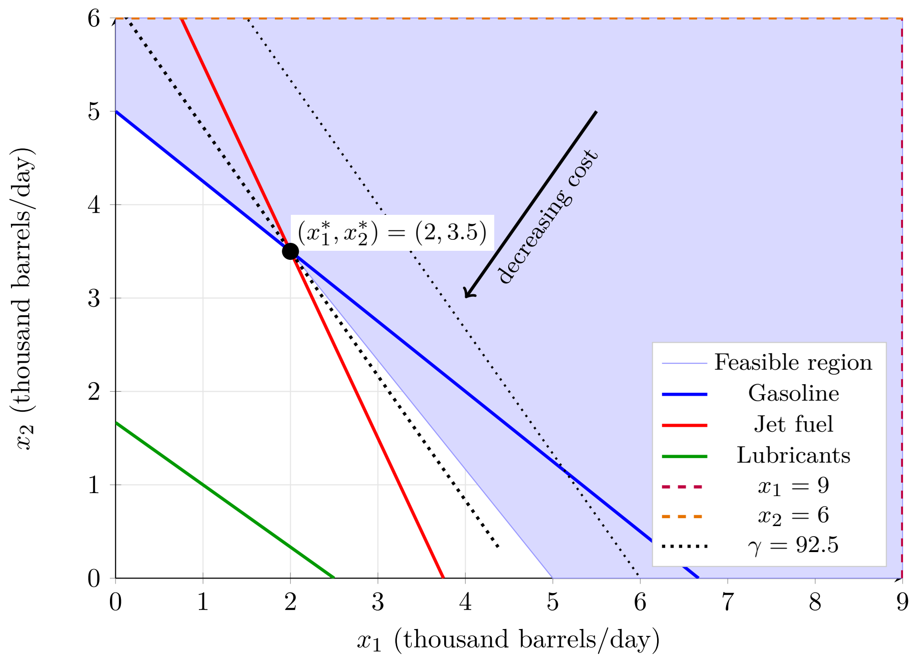


Figure 1: TCP graph

Linear Programming Definition

The general form of a linear program is given in Equation .

$$
\begin{aligned}
\min_x/\max_x \quad & \zeta = c_1x_1 + c_2x_2 + \cdots + c_nx_n + c_0 && (7)\\
\text{s.t.} \quad & b_1^{\mathrm{L}} \le a_{1,1}x_1 + a_{1,2}x_2 + \cdots + a_{1,n}x_n \le b_1^{\mathrm{U}} && (8)\\
& \vdots && (9)\\
& b_m^{\mathrm{L}} \le a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,n}x_n \le b_m^{\mathrm{U}} && (10)\\
& l_1 \le x_1 \le u_1 && (11)\\
& \vdots && (12)\\
& l_n \le x_n \le u_n. && (13)
\end{aligned}
$$

Definitions for solutions:

- The variables $x_1, x_2, \ldots, x_n$ are called *decision variables*, and a, b^{L}, b^{U}, c, l, and u are given *parameters*.
- A solution $x = (x_1, x_2, \ldots, x_n)$ is a *feasible solution* if it satisfies all constraints. A solution is an *infeasible solution* if it violates any constraint.
- The set of all feasible solutions is called the *feasible region* or *feasible set*.
- l_i and u_i are called the lower bound and upper bound on variable x_i, respectively.
- If $l_i = -\infty$ and $u_i = +\infty$, then x_i is a free or unrestricted variable.

Definitions for constraints:

- The symbol 's. t.' is short for 'subject to' or 'such that', which starts the list of constraints.
- b_j^{L} and b_j^{U} are called the lower bound and upper bound on constraint j, respectively.
- If $b_j^{\mathrm{L}} = b_j^{\mathrm{U}}$, the constraint is said to be an equality constraint, and can be represented using $=$ sign.
- A constraint can be represented with only one bound. In this case, the other bound is either $b_j^{\mathrm{L}} = -\infty$ or $b_j^{\mathrm{U}} = +\infty$

Definitions for objectives:

- The function (1) is the *objective function*.

- ζ represents the value of the objective function, called *objective value*. For a given solution $x = (x_1, x_2, \ldots, x_n)$, we use $\zeta(x)$ to denote the objective value of x: $\zeta(x) = c_1x_1 + c_2x_2 + \cdots + c_nx_n + c_0$
- The symbol min or max indicates that we want to minimize or maximize the objective value ζ.
- A solution x^* is *optimal* if it is feasible and $\zeta(x^*) \leq \zeta(x')$ for any other feasible solution x' (for minimization).
- The objective value of the optimal solution is called *optimal value* and is denoted by $\zeta(x^*)$ or ζ^*.

An LP has three possible outcomes:

- LP is *optimal*, which means that there exists an optimal solution, either uniquely or among infinitely many others.
- LP is *infeasible*, which means that a feasible solution does not exist.
- LP is *unbounded*, which means that the objective value goes to infinity (for maximization). In particular, for any real number K, there always exists a feasible solution x such that $\zeta(x) \geq K$.